

Application Firewall

Definition

An **Application Firewall** is a type of firewall that monitors and controls traffic at the **application layer (Layer 7)** of the OSI model. It specifically protects web applications by filtering malicious HTTP/HTTPS requests.

How it Works

- Inspects incoming and outgoing web traffic
- Detects attacks like:
 - SQL Injection
 - Cross-Site Scripting (XSS)
- Blocks harmful requests before they reach the application

Real-World Example

Imagine you are using an online shopping website like Amazon.

- A hacker tries to inject malicious code into the login form
- The **Application Firewall (WAF)** detects unusual input patterns
- It blocks the request before it reaches the server

Result: Your account and data stay safe

Key Features

- Protects web apps specifically
- Understands user inputs and behavior
- Works with HTTP/HTTPS traffic

Next-Generation Firewall (NGFW)

Definition

A **Next-Generation Firewall (NGFW)** is an advanced firewall that goes beyond traditional firewalls by combining multiple security features into one system.

How it Works

NGFW includes:

- Deep Packet Inspection (DPI)

- Intrusion Prevention System (IPS)
- Application awareness
- User identity tracking

It can identify not just **IP addresses**, but also:

- Which app is being used (e.g., social media, streaming)
- Who is using it

Real-World Example

In a company like Infosys:

- Employees are connected to the company network
- NGFW monitors all traffic
- It can:
 - Block access to unsafe websites
 - Allow only work-related apps
 - Detect malware or suspicious behavior

Result: Company network stays secure and controlled

Key Features

- Combines firewall + antivirus + IPS
- Controls applications, not just ports
- Provides real-time threat intelligence

Analyze HTTP/HTTPS Security Issues like Data Manipulation and Hijacking

HTTP Issues (Not Secure)

HTTP does not encrypt data → vulnerable to attacks.

1. Data Manipulation

- Attacker intercepts data and modifies it
- Example: Changing transaction amount

2. Session Hijacking

- Attacker steals session ID
- Gains unauthorized access

3. Man-in-the-Middle (MITM)

- Attacker sits between user and server
 - Reads or alters communication
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HTTPS (Secure Version)

HTTPS uses encryption via SSL/TLS.

Security Advantages

- Data Encryption
- Authentication (server identity)
- Data Integrity

Still Possible Issues

- Fake certificates (phishing sites)
 - SSL stripping attacks
 - Weak encryption
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Example

When using public Wi-Fi:

- HTTP → attacker can read passwords
 - HTTPS → data is encrypted and safe
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Q3. Explain SSL/TLS Handshake Process with Diagram

SSL/TLS Handshake

It is a process that establishes a **secure connection** between client and server.

Steps

1. **Client Hello**
 - Client sends supported encryption methods
2. **Server Hello**
 - Server selects encryption method

- Sends digital certificate

3. **Certificate Verification**

- Client verifies server identity

4. **Key Exchange**

- Session key is generated

5. **Secure Connection Established**

- Data is encrypted using session key

Case Study: Firewall & Network Architecture

Problem

A company faced a data breach despite having:

- Packet filtering firewall
- Stateful firewall

Analysis

These firewalls:

- Only check IP, port, protocol
- Do NOT inspect application data

Attackers used:

- Application-layer attack (e.g., SQL injection)
- Which bypassed basic firewall rules

Weakness

- No deep packet inspection
- No application awareness
- No intrusion detection

Improvements Suggested

1. **Use NGFW**

- Detect advanced threats

- Monitor application behavior
 - 2. **Deploy Application Firewall (WAF)**
 - Protect web applications
 - 3. **Enable IDS/IPS**
 - Detect and prevent intrusions
 - 4. **Regular Updates & Patching**
 - 5. **Network Segmentation**
 - Limit attacker movement
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Protocol Analysis: Packet Filtering

How it Works

- Checks:
 - Source IP
 - Destination IP
 - Port number
 - Protocol

Limitation

- Cannot detect:
 - Malware
 - Application attacks
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Attack Analysis: HTTP Data Manipulation & Lack of Security

Attack

- Attacker intercepts HTTP request
- Modifies data (e.g., price change)

Cause

- No encryption (HTTP)
- No integrity check

Prevention Measures

- Use HTTPS (TLS encryption)
- Use secure cookies
- Implement authentication
- Use firewalls (WAF + NGFW)
- Enable HSTS (forces HTTPS)